

SQL Milestone 1

Analysis - I

1. To simplify its financial reports, Amazon India needs to standardize payment values. Round the average payment values to integer (no decimal) for each payment type and display the results sorted in ascending order.
 - o **Output:** payment_type, rounded_avg_payment

Code Snippet:

```
--Q1

select payment_type,
round(avg(payment_value)) as rounded_avg_payment
from payments
group by payment_type
order by rounded_avg_payment asc
```

Output:

	payment_type character varying	rounded_avg_payment numeric
1	not_defined	0
2	voucher	66
3	debit_card	143
4	boleto	145
5	credit_card	163

Recommendation:

The output shows that credit card payments have the highest average payment value (163). This indicates that customers prefer using credit cards for higher-value purchases on Amazon. We can recommend that Amazon India introduce more credit card offers to attract customers and encourage high-value purchases. Credit cards are likely preferred for larger payments because customers can convert them into easy monthly instalments (EMIs).

2. **To refine its payment strategy, Amazon India wants to know the distribution of orders by payment type.** Calculate the percentage of total orders for each payment type, rounded to one decimal place, and display them in descending order
 - **Output:** payment_type, percentage_orders

Code Snippet:

```
select payment_type,
       round(
         (count( distinct order_id)::numeric /
          (select count( distinct order_id) from payments)) * 100
         , 1)
       as percentage_orders
from payments
group by payment_type
order by percentage_orders desc
```

Output:

Data Output			Messages	Notifications
	payment_type	percentage_orders		
	character varying	numeric		
1	credit_card	73.9		
2	boleto	19.0		
3	voucher	5.6		
4	debit_card	1.5		
5	not_defined	0.0		

Recommendation:

The analysis shows that credit cards are the most preferred payment method among customers, accounting for 73.9% of total orders. This indicates a strong customer inclination toward card-based payments, especially for higher-value purchases. Amazon India can leverage this behaviour by introducing targeted credit card offers, cashback deals, and EMI options to further drive sales and increase average order value.

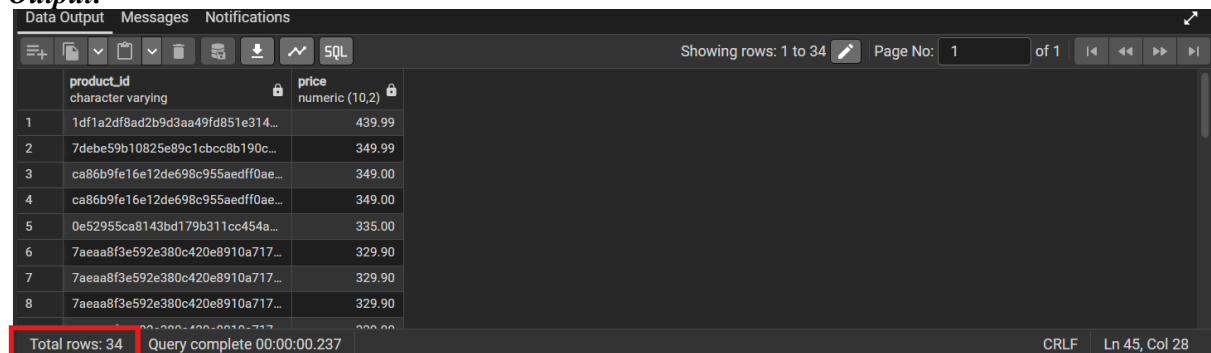
3. **Amazon India seeks to create targeted promotions for products within specific price ranges.** Identify all products priced between 100 and 500 BRL that contain the word 'Smart' in their name. Display these products, sorted by price in descending order.

- **Output:** product_id, price

Code Snippet:

```
SELECT
    p.product_id,
    o.price
FROM product AS p
INNER JOIN order_items AS o
    ON o.product_id = p.product_id
WHERE o.price BETWEEN 100 AND 500
    AND p.product_category_name ILIKE '%Smart%'
ORDER BY o.price DESC
```

Output:



	product_id character varying	price numeric (10,2)
1	1df1a2df8ad2b9d3aa49fd851e314...	439.99
2	7debe59b10825e89c1cbcc8b190c...	349.99
3	ca86b9fe16e12de698c955aedff0ae...	349.00
4	ca86b9fe16e12de698c955aedff0ae...	349.00
5	0e52955ca8143bd179b311cc454a...	335.00
6	7aeaa8f3e592e380c420e8910a717...	329.90
7	7aeaa8f3e592e380c420e8910a717...	329.90
8	7aeaa8f3e592e380c420e8910a717...	329.90

Total rows: 34 Query complete 00:00:00.237 CRLF Ln 45, Col 28

Recommendation:

The presence of 34 “Smart” products within the mid-price range indicates strong demand for affordable smart devices among Indian customers. This suggests that consumers are actively looking for value-for-money smart products. Amazon India can introduce targeted campaigns such as “Smart Picks Under ₹X” to attract price-sensitive customers and increase sales. This segment also provides opportunities for upselling premium smart devices, accessories, and extended warranties.

4. To identify seasonal sales patterns, Amazon India needs to focus on the most successful months. Determine the top 3 months with the highest total sales value, rounded to the nearest integer.

○ **Output:** month, total_sales

Code Snippet:

```
select date_trunc('month', o.order_purchase_timestamp) as month,
round(sum(op.price)) as total_sales
from orders as o
join
order_items as op
on
o.order_id=op.order_id
group by month
order by total_sales desc
limit 3
```

Output:

Data Output			Messages	Notifications
	month timestamp without time zone	payment_value numeric		
1	2017-11-01 00:00:00	1194883		
2	2018-04-01 00:00:00	1160785		
3	2018-03-01 00:00:00	1159652		

Recommendation:

The analysis reveals a clear seasonal pattern, with March and April recording the highest payment values. These months appear to be peak sales periods, possibly driven by increased travel and spending during the Carnival season. This suggests that customer purchasing activity rises significantly during festive and holiday periods.

5. **Amazon India is interested in product categories with significant price variations.** Find categories where the difference between the maximum and minimum product prices is greater than 500 BRL.

○ **Output:** product_category_name, price_difference

Code Snippet:

```
select
p.product_category_name as product_category_name,
max(o.price)-min(o.price) as price_difference
from product as p
inner join
order_items as o
on
p.product_id=o.product_id
where p.product_category_name is not null
and
o.price is not null
group by p.product_category_name
having max(o.price)-min(o.price)>500
```

Output:

Data Output Messages Notifications		
SQL		
	product_category_name character varying	price_difference numeric
1	climatizacao	1588.10
2	livros_importados	730.01
3	ferramentas_jardim	3923.65
4	dvds_blu_ray	1411.10
5	cine_foto	867.19
6	beleza_saude	3122.80
7	livros_interesse_geral	893.90
8	tablets_impressao_imagem	875.09
Total rows: 56		Query complete 00:00:00.275

Recommendation:

1. Multiple Customer Segments

- These categories serve budget, mid-range, and premium customers.
- Opportunity to target different income groups within the same category.

2. Premium Product Opportunity

- High price variation indicates the presence of expensive products.
- Amazon India can promote premium items.
- EMI options can encourage high-value purchases.

3. Pricing Strategy Improvement

- Large price gaps allow better pricing flexibility.
- Amazon can use dynamic pricing and competitive pricing strategies.

4. Promotion & Upselling Opportunity

- Wide price ranges make discounts more attractive.
- Opportunity to create “Best under ₹X” sections.
- Bundle offers can increase average order value.

Conclusion

The 56 categories with high price variation provide strong opportunities for:

- Multi-segment targeting
- Premium positioning
- Better pricing strategies
- Increased sales and profitability

6. To enhance the customer experience, Amazon India wants to find which payment types have the most consistent transaction amounts. Identify the payment types with the least variance in transaction amounts, sorting by the smallest standard deviation first.

○ **Output:** payment_type, std_deviation

Code Snippet:

```
select payment_type,  
round(stddev(payment_value),2) as std_deviation  
from payments  
group by payment_type  
order by stddev(payment_value) asc
```

Output:

	payment_type	std_deviation
	character varying	numeric
1	not_defined	0.00
2	voucher	115.52
3	boleto	213.58
4	credit_card	222.12
5	debit_card	245.79

Recommendation:

Voucher payments show consistent transaction amounts, which means customers usually spend within a fixed range when using vouchers. In contrast, credit and debit card payments show higher variation, indicating they are used for both small and high-value purchases. Therefore, Amazon India can introduce EMI options and cashback offers on card payments to encourage more high-value purchases while continuing targeted voucher campaigns for stable spending segments.

7. **Amazon India wants to identify products that may have incomplete name in order to fix it from their end.** Retrieve the list of products where the product category name is missing or contains only a single character.

○ **Output:** product_id, product_category_name

Code Snippet:

```
select product_id, product_category_name
from product
where product_category_name is null
or length(trim(product_category_name)) = 1
```

Output:

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 614

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	product_id [PK] character varying	product_category_name character varying
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdbbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]
8	794de06c32a626a5692ff50e4985d3...	[null]
Total rows: 614 Query complete 00:00:00.174		

CRLF Ln 156, Col 4

Recommendation:

The presence of missing or single-character product category names indicates data quality issues in the product catalogue. Incorrect categorization can negatively impact search results, product recommendations, and sales analysis. Amazon India should clean and standardize these category names to improve product visibility, customer experience, and reporting accuracy.

Analysis - II

1. **Amazon India wants to understand which payment types are most popular across different order value segments (e.g., low, medium, high).** Segment order values into three ranges: orders less than 200 BRL, between 200 and 1000 BRL, and over 1000 BRL. Calculate the count of each payment type within these ranges and display the results in descending order of count

- Output: order_value_segment, payment_type, count

Code Snippet:

```
select s.order_value_segment as order_value_segment,
s.payment_type as payment_type,
count(*) as count
from
(
select
case when payment_value < 200 then 'low'
when payment_value >= 200 and payment_value < 1000 then 'medium'
else 'high'
end
as order_value_segment,
payment_type
from payments
) s
group by order_value_segment, payment_type
order by count desc
```

Output:

Data Output Messages Notifications			
SQL			
	order_value_segment text	payment_type character varying	count bigint
1	low	credit_card	60548
2	low	boleto	16444
3	medium	credit_card	15303
4	low	voucher	5476
5	medium	boleto	3162
6	low	debit_card	1287
7	high	credit_card	944
8	medium	voucher	286
9	medium	debit_card	227
Total rows: 13		Query complete 00:00:00.162	

Recommendation:

Credit card payments are the most popular across all order value segments — low, medium, and high. This supports the earlier finding that credit cards have a higher standard deviation, indicating they are used for a wide range of transaction amounts. This suggests that customers prefer credit cards for both small and high-value purchases. Amazon India can leverage this behaviour by offering EMI options and targeted credit card promotions to drive higher-value transactions.

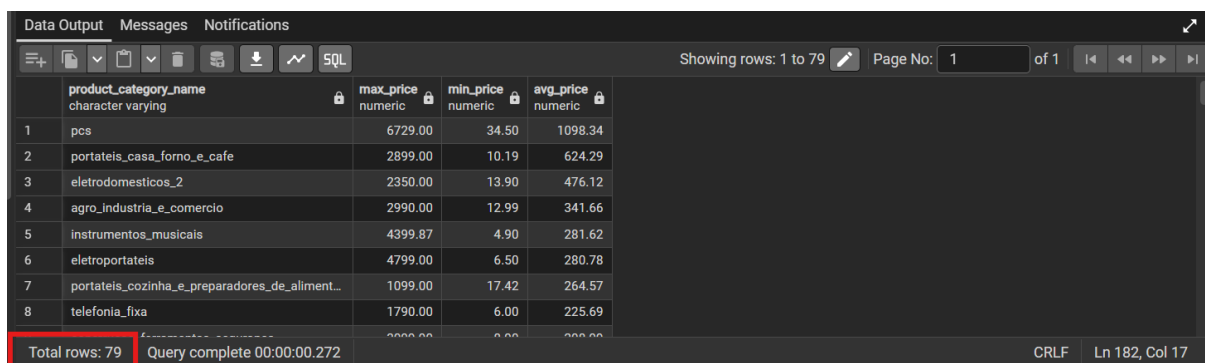
2. **Amazon India wants to analyse the price range and average price for each product category.** Calculate the minimum, maximum, and average price for each category, and list them in descending order by the average price.

○ **Output:** product_category_name, min_price, max_price, avg_price

Code Snippet:

```
select p.product_category_name as product_category_name,
max(o.price) as max_price,
min(o.price) as min_price,
round(avg(o.price),2) as avg_price
from order_items as o
join
product as p
on
o.product_id=p.product_id
group by p.product_category_name
order by avg_price desc
```

Output:



The screenshot shows a database interface with a table of query results. The table has four columns: product_category_name, max_price, min_price, and avg_price. The data is sorted by avg_price in descending order. The first row shows 'pcs' with a max price of 6729.00, min price of 34.50, and avg price of 1098.34. The last row shows 'telefonos_fixos' with a max price of 1790.00, min price of 6.00, and avg price of 225.69. The interface also shows 'Total rows: 79' and 'Query complete 00:00:00.272'.

	product_category_name character varying	max_price numeric	min_price numeric	avg_price numeric
1	pcs	6729.00	34.50	1098.34
2	portateis_casa_forno_e_cafe	2899.00	10.19	624.29
3	eletrodomesticos_2	2350.00	13.90	476.12
4	agro_industria_e_comercio	2990.00	12.99	341.66
5	instrumentos_musicais	4399.87	4.90	281.62
6	eletroportateis	4799.00	6.50	280.78
7	portateis_cozinha_e_preparadores_de_aliment...	1099.00	17.42	264.57
8	telefonos_fixos	1790.00	6.00	225.69

Total rows: 79 Query complete 00:00:00.272 CRLF Ln 182, Col 17

Recommendation:

1. High Average Price Categories (Premium Segment)

- Introduce EMI options and extended warranties to encourage high-value purchases.
- Focus marketing efforts on premium positioning and high-income customer segments.

2. Categories with Wide Price Range (Mixed Segment)

- Create price-based filters such as “Under ₹500” or “Under ₹1000” to improve customer navigation.
- Use upselling strategies to promote higher-priced products within the same category.

3. Low Average Price Categories (Budget Segment)

- Run discount campaigns and bundle offers to increase sales volume.
- Focus on competitive pricing to attract price-sensitive customers.

Summary

- Use category-level pricing insights to apply different strategies instead of a single pricing approach.
- Align promotions, marketing, and payment options based on the pricing behaviour of each category

3. **Amazon India wants to identify the customers who have placed multiple orders over time.** Find all customers with more than one order, and display their customer unique IDs along with the total number of orders they have placed.
- **Output:** customer_unique_id, total_orders

Code Snippet:

```
select c.customer_unique_id as customer_unique_id,
count(o.order_id) as total_orders
from customers as c
join
orders as o
on
c.customer_id=o.customer_id
group by c.customer_unique_id
having count(o.order_id)>1
order by total_orders desc
```

Output:

Data Output		Messages	Notifications
			
	customer_unique_id character varying	total_orders bigint	
1	a91e80fbe80ddc07de66a5cf927029...	16	
2	a6168cd79131e64acef92e3c74d6cc...	16	
3	363f980585bf04c1a88fdb986011c5...	16	
4	cbd0350d4ccba9772e8e768d4a4a5...	16	
5	417b909c0962b2610f1cfeb1c14789...	16	
6	5f94af52aef02c968a2e0f01f430864e	16	
7	1b6d29725255a77667a8c639eeb4c...	16	
8	e4bbcc533fdf3917c56dea2c43bf20...	16	
9	000-1000-55067334447-0-1-025-	16	
Total rows: 3140		Query complete 00:00:01.398	

Recommendation:

The analysis identifies customers who have placed more than one order, indicating repeat purchase behaviour. These customers are valuable because they show trust and continued engagement with the Amazon platform. Amazon India can target these repeat customers with personalized offers, loyalty rewards, and exclusive discounts to strengthen retention. Additionally, satisfied repeat customers can act as passive brand advocates by recommending the platform to others, helping drive organic growth.

- Introduce loyalty programs and personalized offers for repeat customers.
- Provide exclusive early access deals to increase retention and lifetime value.
- Encourage referrals to turn loyal customers into brand advocates.

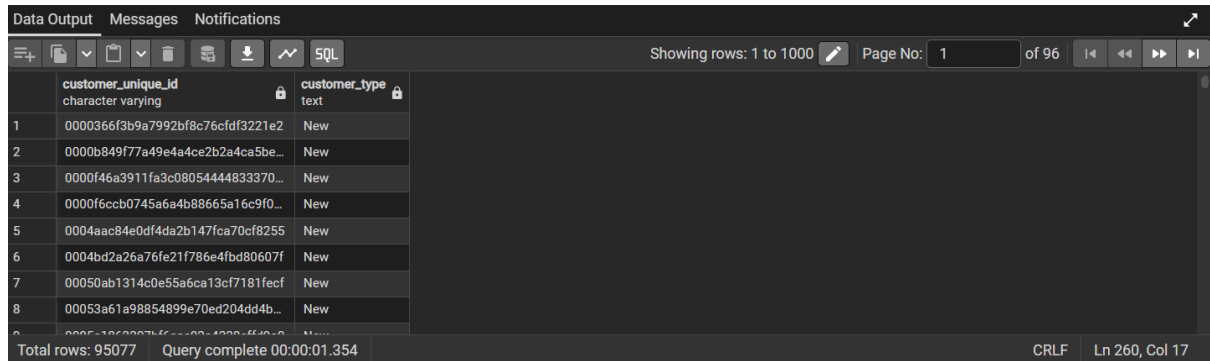
4. **Amazon India wants to categorize customers into different types ('New – order qty. = 1'; 'Returning' –order qty. 2 to 4; 'Loyal' – order qty. >4) based on their purchase history.** Use a temporary table to define these categories and join it with the customers table to update and display the customer types.

○ **Output:** customer_unique_id, customer_type

Code Snippet:

```
with customers as
(
select c.customer_unique_id as customer_unique_id,
count(o.order_id) as total_orders ,
case when count(o.order_id)=1 then 'New'
when count(o.order_id) between 2 and 4 then 'Returning'
when count(o.order_id)>4 then 'Loyal'
end as
customer_type
from orders as o
join
customers as c
on c.customer_id=o.customer_id
group by customer_unique_id
)
select customer_unique_id, customer_type from customers
```

Output:



	customer_unique_id character varying	customer_type text
1	0000366f3b9a7992bf8c76cfd3221e2	New
2	0000b849f77a49e4a4ce2b2a4ca5be...	New
3	0000f46a3911fa3c08054444833370...	New
4	0000f6ccb0745a6a4b88665a16c9f0...	New
5	0004aac84e0df4da2b147fca70cf8255	New
6	0004bd2a26a76fe21f786e4fbd80607f	New
7	00050ab1314c0e55a6ca13cf7181fecf	New
8	00053a61a98854899e70ed204dd4b...	New

Total rows: 95077 Query complete 00:00:01.354 CRLF Ln 260, Col 17

Recommendation:

Reward Loyal Customers

Offer exclusive discounts, early access to sales, and loyalty rewards to customers with more than 4 orders to increase retention and lifetime value.

Convert Returning Customers into Loyal Customers

Provide personalized offers, targeted promotions, and limited-time incentives to customers with 2–4 orders to encourage more frequent purchases.

Engage and Retain New Customers

Introduce welcome offers, first-repeat purchase discounts, and onboarding campaigns to motivate customers with only 1 order to shop again.

5. **Amazon India wants to know which product categories generate the most revenue.** Use joins between the tables to calculate the total revenue for each product category. Display the top 5 categories.

○ **Output:** product_category_name, total_revenue

Code Snippet:

```
select p.product_category_name as product_category_name,
sum(o.price) as total_revenue
from product as p
join
order_items as o
on
p.product_id=o.product_id
group by p.product_category_name
order by total_revenue desc
Limit 5
```

Output:

Data Output			Messages	Notifications
	product_category_name character varying	total_revenue numeric		
1	beleza_saude	1257865.34		
2	relogios_presentes	1203060.32		
3	cama_mesa_banho	1032268.59		
4	esporte_lazer	985881.10		
5	informatica_acessorios	910605.07		

Recommendation:

Focus Marketing on Top Revenue Categories

Increase advertising, promotions, and visibility for the top 5 revenue-generating categories to further boost sales and maximize profit.

Ensure Strong Inventory Management

Maintain sufficient stock levels in these high-performing categories to avoid stockouts and prevent revenue loss.

Introduce Premium and Bundle Offers

Launch combo deals, cross-selling, and premium product promotions within these categories to increase average order value and overall revenue.

Analysis - III

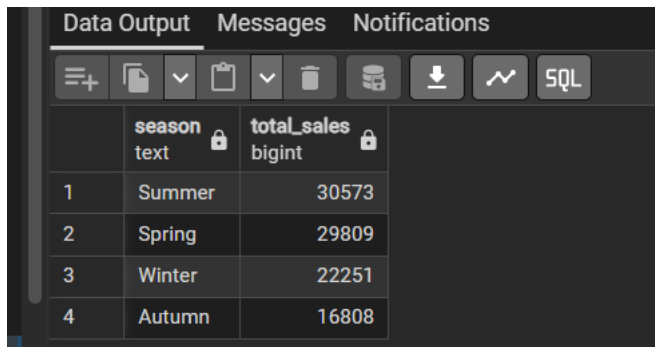
1. **The marketing team wants to compare the total sales between different seasons.** Use a subquery to calculate total sales for each season (Spring, Summer, Autumn, Winter) based on order purchase dates, and display the results. Spring is in the months of March, April and May. Summer is from June to August and Autumn is between September and November and rest months are Winter.

- **Output:** season, total_sales

Code Snippet:

```
SELECT
  CASE
    WHEN month_number IN (3, 4, 5) THEN 'Spring'
    WHEN month_number IN (6, 7, 8) THEN 'Summer'
    WHEN month_number IN (9, 10, 11) THEN 'Autumn'
    ELSE 'Winter'
  END AS season,
  sum(price) AS total_sales
FROM (
  SELECT
    op.price,
    EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month_number
  FROM orders as o
  join order_items as op
  on o.order_id=op.order_id
) sub
GROUP BY season
ORDER BY total_sales desc
```

Output:



	season text	total_sales bigint
1	Summer	30573
2	Spring	29809
3	Winter	22251
4	Autumn	16808

Recommendation:

Summer is a high-sales season. In India, summer includes school holidays, outdoor activities, travel, and events like IPL. During this time, children and young adults spend more time on sports, swimming, travel, and entertainment. Families also purchase seasonal products such as sunglasses, sports goods, cooling appliances, and TVs. This increases overall demand.

1. **Increase Marketing During Summer**
Invest more in advertising and promotions during summer months, especially around school holidays and IPL season, to capture higher customer spending.

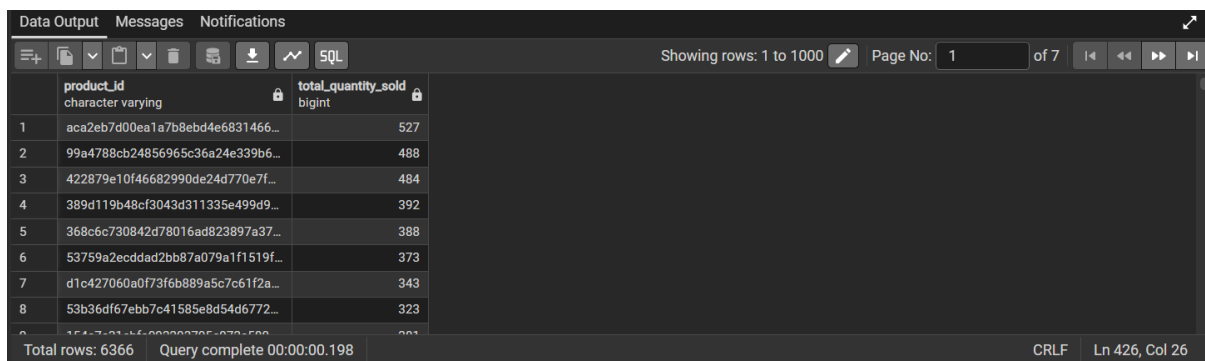
2. **Promote Seasonal Products**
Highlight products such as sunglasses, sports equipment, swimming gear, air coolers, and TVs through targeted campaigns and homepage banners.
3. **Plan Inventory in Advance**
Ensure sufficient stock of high-demand summer products to avoid stock shortages and maximize revenue during peak season.

2. **The inventory team is interested in identifying products that have sales volumes above the overall average.** Write a query that uses a subquery to filter products with a total quantity sold above the average quantity.
 - **Output:** product_id, total_quantity_sold

Code Snippet:

```
select product_id, count(*) as total_quantity_sold from order_items
group by product_id
having count(*) >
(
select avg(total_quantity_sold) from
(
select product_id, count(*) as total_quantity_sold
from order_items
group by product_id
) as avg_table
)
order by total_quantity_sold desc
```

Output:



The screenshot shows a database interface with a table of query results. The table has two columns: 'product_id' (character varying) and 'total_quantity_sold' (bigint). The results are sorted in descending order of total quantity sold. The first 8 rows are visible, showing product IDs and their corresponding total quantities sold.

	product_id	total_quantity_sold
1	aca2eb7d00ea1a7b8ebd4e6831466...	527
2	99a4788cb24856965c36a24e339b6...	488
3	422879e10f46682990de24d770e7f...	484
4	389d119b48cf3043d311335e499d9...	392
5	368c6c730842d78016ad823897a37...	388
6	53759a2ecddad2bb87a079a1f1519f...	373
7	d1c427060a0f73f6b889a5c7c61f2a...	343
8	53b36df67ebb7c41585e8d54d6772...	323

At the bottom of the window, it says 'Total rows: 6366' and 'Query complete 00:00:00.198'. The status bar at the bottom right shows 'CRLF' and 'Ln 426, Col 26'.

Recommendation:

The products with total quantity sold above the overall average represent high-demand and fast-moving items. These products are frequently purchased and contribute significantly to overall sales volume.

1. **Prioritize Inventory Management**
Ensure these high-demand products are always well stocked to avoid stockouts and revenue loss.
2. **Optimize Delivery and Logistics**
Improve warehouse placement and delivery speed for these products to maintain customer satisfaction.
3. **Leverage for Promotions**
Highlight these popular products in “Best Sellers” or “Trending Now” sections to further increase sales.

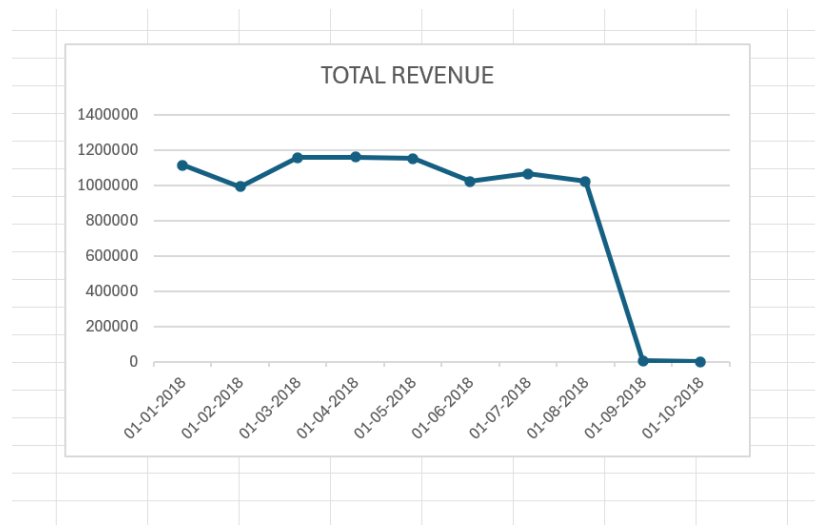
The above-average selling products indicate strong and consistent customer demand. Amazon India should prioritize inventory planning, faster delivery logistics, and promotional visibility for these products to maximize revenue and enhance customer experience.

- **Output:** Month, total_revenue

```
with monthly_sales as
(
select order_id,
date_trunc('month', order_purchase_timestamp)::date AS month
from orders
WHERE order_purchase_timestamp >= '2018-01-01'
AND order_purchase_timestamp < '2019-01-01'
)
select monthly_sales.month,
sum(p.payment_value) as total_revenue
from monthly_sales
join
payments as p
on
p.order_id=monthly_sales.order_id
group by monthly_sales.month
order by monthly_sales.month asc
```

Data Output		Messages	Notifications						
									SQL
	month date		total_revenue numeric						
1	2018-01-01		1115004.18						
2	2018-02-01		992463.34						
3	2018-03-01		1159652.12						
4	2018-04-01		1160785.48						
5	2018-05-01		1153982.15						
6	2018-06-01		1023880.50						
7	2018-07-01		1066540.75						
8	2018-08-01		1022425.32						
9	2018-09-01		1122554.14						
Total rows: 10		Query complete 00:00:00.151							

Revenue data month by month:



Recommendation:

1. Replicate Successful Strategies from Peak Months (March–May)

- Analyse which products, promotions, and marketing campaigns performed best during March to May.
- Apply these successful strategies during slower months to help increase overall revenue.

2. Strengthen Mid-Year Performance (June–August)

- Introduce targeted discounts, bundle offers, or limited-time promotions to stimulate demand.
- Increase marketing efforts during these months to maintain consistent sales levels.

3. Plan Proactively for High-Sales Periods

- Ensure sufficient inventory and operational readiness before peak sales months.
- Increase advertising and promotional activities in advance to maximize revenue opportunities.

4. Improve Revenue Stability Throughout the Year

- Implement customer loyalty programs to encourage repeat purchases.
- Use targeted communication strategies, such as email marketing and personalized offers, to maintain customer engagement year-round.

4. **A loyalty program is being designed for Amazon India.** Create a segmentation based on purchase frequency: 'Occasional' for customers with 1-2 orders, 'Regular' for 3-5 orders, and 'Loyal' for more than 5 orders. Use a CTE to classify customers and their count and generate a chart in Excel to show the proportion of each segment.

○ **Output:** customer_type, count

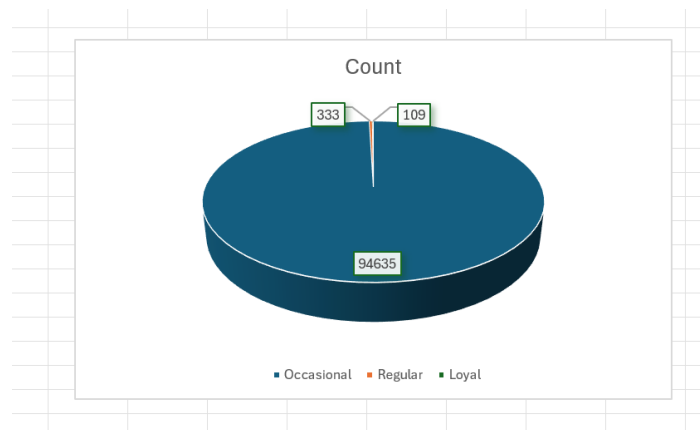
Code Snippet:

```
with customer_classification as
(
select c.customer_unique_id,
case
when count(o.order_id) between 1 and 2 then 'Occasional'
when count(o.order_id) between 3 and 5 then 'Regular'
when count(o.order_id)>5 then 'Loyal'
end as customer_type,
count(o.order_id) as count_of_orders from customers as c
join
orders as o
on
c.customer_id=o.customer_id
group by c.customer_unique_id
)
select customer_type, count(*) as count
from customer_classification
group by customer_type
```

Output:

Data Output		Messages	Notifications
	customer_type	count	
	text	bigint	
1	Occasional	94635	
2	Regular	333	
3	Loyal	109	

Customer type by count:



Recommendation:

1. Focus on Converting Occasional Customers

- Offer incentives after the first or second purchase (discount codes, reward points, free delivery).
- Send personalized follow-up communication to encourage the next purchase.

Even a small improvement in conversion from Occasional to Regular can significantly increase revenue.

2. Nurture Regular Customers into Loyal Customers

Regular customers are close to becoming loyal:

- Introduce tier-based rewards (e.g., Silver, Gold, Platinum levels).
- Offer exclusive early access to deals or special benefits after 3–5 purchases.

This group is the easiest to grow into long-term loyal customers.

3. Protect and Reward Loyal Customers

Although they are few, loyal customers are very valuable:

- Provide exclusive perks such as priority support, special discounts, or birthday rewards.
- Recognize them with personalized communication to strengthen emotional connection.

Retaining loyal customers is usually cheaper than acquiring new ones.

4. Improve First-Purchase Experience

Since most customers are occasional:

- Ensure fast delivery, easy returns, and responsive customer support.
- Make the checkout and payment process smooth and simple.
- A strong first experience increases the probability of repeat purchases.

5. **Amazon wants to identify high-value customers to target for an exclusive rewards program.** You are required to rank customers based on their average order value (avg_order_value) to find the top 20 customers.
- **Output: customer_id, avg_order_value, customer_rank**

Code Snippet:

```
with order_totals as
(
select order_id,
sum(payment_value) as order_total
from payments
group by order_id
),
order_value as
(
select o.customer_id, round(avg(ot.order_total),2) as average_order_value,
dense_rank() over (order by round(avg(ot.order_total),2) desc)
as customer_rank
from order_totals as ot
join orders as o
on
o.order_id=ot.order_id
group by o.customer_id
)
select customer_id, average_order_value,
customer_rank
from order_value
where customer_rank<=20
```

Output:

Data Output				Messages	Notifications
				SQL	
	customer_id character varying	average_order_value numeric	customer_rank bigint		
1	1617b1357756262bfa56ab541c47b...	13664.08	1		
2	ec5b2ba62e574342386871631fafd...	7274.88	2		
3	c6e2731c5b391845f6800c97401a4...	6929.31	3		
4	f48d464a0baaea338cb25f816991a...	6922.21	4		
5	3fd6777bbce08a352fddd04e4a7cc...	6726.66	5		
6	05455dfa7cd02f13d132aa7a6a972...	6081.54	6		
7	df55c14d1476a9a3467f131269c24...	4950.34	7		
8	e0a2412720e9ea4f26c1ac985f6a7...	4809.44	8		
9	24bbf5fd2f2e1b359ee7de94defc4a...	4764.34	9		
10	3d979689f636322c62418b6346b1c...	4681.78	10		
11	1afc82cd60e303ef09b4ef9837c950...	4513.32	11		
12	cc803a2c412833101651d3f90ca7d...	4445.50	12		
13	926b6a6fb8b6081e00b335edaf578...	4194.76	13		
14	35a413c7ca3c69756cb75867d6311...	4175.26	14		
15	e9b0d0eb3015ef1c9ce6cf5b9dcbec...	4163.51	15		
16	3be2c536886b2ea4668eced3a80dd...	4042.74	16		
17	eb7a157e8da9c488cd4ddc48711f1...	4034.44	17		
18	c6695e3b1e48680db36b487419fb0...	4016.91	18		
19	31e83c01fce824d0ff786fcd48dad0...	3979.55	19		
20	addc91fdf9c2b3045497b57fc710e8...	3826.80	20		

Recommendation:

1. Launch an Exclusive Premium Rewards Tier

- Create a VIP or Elite membership tier specifically for high AOV customers.
- Offer exclusive benefits such as early access to sales, premium customer support, or special discounts.

This will strengthen loyalty and increase retention among high-spending customers.

2. Provide Personalized Experiences

- Offer personalized product recommendations based on purchase history.
- Send tailored offers for high-value or premium products.

High AOV customers are more likely to respond positively to curated experiences.

3. Protect and Retain High-Value Customers

- Assign priority customer support to ensure quick issue resolution.
- Monitor churn risk for these customers and proactively engage them if purchase frequency declines.

Retaining one high-value customer is often more profitable than acquiring several new low-value customers.

4. Encourage Higher Spending Across Other Segments

- Study the buying patterns of top 20 customers.
- Use insights (product categories, pricing range, bundles) to encourage mid-tier customers to increase their average order value.

This helps scale revenue beyond just the top segment.

6. **Amazon wants to analyze sales growth trends for its key products over their lifecycle.** Calculate monthly cumulative sales for each product from the date of its first sale. Use a recursive CTE to compute the cumulative sales (total_sales) for each product month by month.
- Output: product_id, sale_month, and total_sales

Code Snippet:

```
--ANALYSIS -3 Q6
with recursive main_table as (
    select
        oi.product_id,
        TO_CHAR(o.order_purchase_timestamp, 'Month') AS sale_month,
        oi.price,
        row_number() over(partition by oi.product_id order by o.order_purchase_timestamp)
        as row_num
    from order_items as oi
    join orders as o
    using (order_id)
),
table_cte as (
    select
        product_id,
        sale_month,
        price,
        price::numeric as cumulative_sales,
        row_num -- Base query
    from main_table
    where row_num = 1
    union all
    select
        m.product_id,
        m.sale_month,
        m.price,
        m.price + t.cumulative_sales,
        m.row_num -- Recursive Query
    from main_table as m
    join table_cte as t
    using (product_id)
    where m.row_num = t.row_num + 1
)
```

```
select
    product_id,
    sale_month,
    sum(cumulative_sales) over (partition by product_id order by sale_month)
    as total_sales
from table_cte
order by product_id, row_num
```

Output:

Data Output

Messages

Notifications

SQL

	product_id character varying	sale_month text	total_sales numeric	
1	00066f42aeeb9f3007548bb9d3f33c38	May	101.65	
2	00088930e925c41fd95ebfe695fd2655	December	129.90	
3	0009406fd7479715e4bef61dd91f2462	December	229.00	
4	000b8f95fcb9e0096488278317764d...	August	176.70	
5	000b8f95fcb9e0096488278317764d...	August	176.70	
6	000d9be29b5207b54e86aa1b1ac548...	April	199.00	
7	0011c512eb256aa0dbbb544d8dffcf6e	December	52.00	
8	00126f27c813603687e6ce486d909d...	September	747.00	
9	00126f27c813603687e6ce486d909d...	September	747.00	
10	001795ec6f1b187d37335e1c470476...	October	1750.50	
	001795ec6f1b187d37335e1c470476...	November	1711.60	
Total rows: 112650		Query complete 00:00:30.199		

Recommendation:

This helps us in analysing the lifecycle sales trend of each product by tracking cumulative sales month by month starting from the product's first sale. It allows Amazon to observe how product demand grows over time and identify long-term sales patterns.

7. To understand how different payment methods affect monthly sales growth, Amazon wants to compute the total sales for each payment method and calculate the month-over-month growth rate for the past year (year 2018). Write query to first calculate total monthly sales for each payment method, then compute the percentage change from the previous month.

- Output: payment_type, sale_month, monthly_total, monthly_change.

Code Snippet:

```
with total_sales as
(
  select p.payment_type,
  date_trunc('month', o.order_purchase_timestamp):: date as sale_month,
  sum(p.payment_value) as monthly_sales,
  lag(sum(p.payment_value))
  over(partition by payment_type
  order by date_trunc('month', o.order_purchase_timestamp):: date asc)
  as monthly_change
  from payments as p
  join
  orders as o
  on
  p.order_id=o.order_id
  where order_purchase_timestamp >= '2018-01-01'
  and order_purchase_timestamp < '2019-01-01'
  group by p.payment_type, sale_month
  order by payment_type asc, sale_month asc
)
select
payment_type, sale_month, monthly_sales,
ROUND(
  ((monthly_sales - monthly_change)
  / NULLIF(monthly_change, 0)
  ) * 100
, 2) as monthly_change
from total_sales;
```

Output:

Data Output

Messages

Notifications

SQL

	payment_type character varying	sale_month date	monthly_sales numeric	monthly_change numeric	
1	boleto	2018-01-01	204844.66	[null]	
2	boleto	2018-02-01	183112.72	-10.61	
3	boleto	2018-03-01	191538.02	4.60	
4	boleto	2018-04-01	193547.09	1.05	
5	boleto	2018-05-01	195378.93	0.95	
6	boleto	2018-06-01	153350.28	-21.51	
7	boleto	2018-07-01	198041.24	29.14	
8	boleto	2018-08-01	143805.90	-27.39	
9	credit_card	2018-01-01	868880.38	[null]	
10	credit_card	2018-02-01	778803.00	-10.37	
11	credit_card	2018-03-01	933770.10	19.90	
12	credit_card	2018-04-01	934306.00	0.06	
13	credit_card	2018-05-01	927556.35	-0.72	
14	credit_card	2018-06-01	811508.56	-12.51	
15	credit_card	2018-07-01	803674.49	-0.97	
Total rows: 36		Query complete 00:00:00.287			

Data Output		Messages	Notifications	
				
			SQL	
	payment_type character varying 	sale_month date 	monthly_sales numeric 	monthly_change numeric 
16	credit_card	2018-08-01	797648.89	-0.75
17	debit_card	2018-01-01	11543.55	[null]
18	debit_card	2018-02-01	7469.53	-35.29
19	debit_card	2018-03-01	8375.11	12.12
20	debit_card	2018-04-01	10782.53	28.74
21	debit_card	2018-05-01	9710.74	-9.94
22	debit_card	2018-06-01	35672.62	267.35
23	debit_card	2018-07-01	44866.18	25.77
24	debit_card	2018-08-01	46001.33	2.53
25	not_defined	2018-08-01	0.00	[null]
26	not_defined	2018-09-01	0.00	[null]
27	voucher	2018-01-01	29735.59	[null]
28	voucher	2018-02-01	23078.09	-22.39
29	voucher	2018-03-01	25968.89	12.53
30	voucher	2018-04-01	22149.86	-14.71
31	voucher	2018-05-01	21336.13	-3.67
32	voucher	2018-06-01	23349.04	9.43
33	voucher	2018-07-01	19958.84	-14.52
34	voucher	2018-08-01	34969.20	75.21
35	voucher	2018-09-01	4439.54	-87.30
36	voucher	2018-10-01	589.67	-86.72
Total rows: 36		Query complete 00:00:00.287		

Recommendation:

1. Credit Card (Main Revenue Driver – Most Stable)

Context:

Credit card generates the highest monthly sales and shows relatively stable month-to-month changes compared to other payment methods.

Recommendations:

- Focus on optimizing the credit card checkout process since it contributes the majority of revenue. Even small improvements can significantly impact total sales.
- Monitor and reduce payment failures or declined transactions to maintain steady growth.

2. Boleto (Moderate Sales – Fluctuates Often)

Context:

Boleto shows noticeable monthly ups and downs, meaning customer usage is not very stable.

Recommendations:

- Send reminders before boleto expiration to reduce incomplete or abandoned payments.
- Offer small incentives, such as limited-time discounts, to encourage customers to complete their boleto transactions.

3. Debit Card (Low Sales – Highly Volatile)

Context:

Debit card has lower total sales but experiences large percentage changes from month to month.

Recommendations:

- Investigate what caused large growth months and try to replicate those successful factors.
- Promote debit card as an alternative payment option during marketing campaigns to gradually increase its adoption.

4. Voucher (Campaign-Driven – Highly Unstable)

Context:

Voucher sales increase sharply during certain months and then drop significantly, suggesting they are promotion-driven.

Recommendations:

- Instead of running one large promotion, consider smaller and more consistent voucher campaigns throughout the year.
- Measure whether vouchers are attracting new customers or simply reducing margins to ensure they are profitable.

5. Not Defined (Data Quality Concern)

Context:

The “not defined” category shows zero sales and no meaningful changes, which likely indicates missing or incorrect classification.

Recommendations:

- Clean and properly categorize this data to improve reporting accuracy.
- Implement checks to prevent payment types from being recorded as undefined in the future.

Summary:

Credit card is the most important and stable payment method and should be protected and optimized. Other payment methods show higher volatility and appear more sensitive to promotions or behavioral factors. Understanding and managing these fluctuations can help improve overall monthly sales growth.